

Fewer storm deaths, but heat is still a risk

U.S. storm deaths have fallen relative to population. Recent heat deaths and migration to Phoenix show why that is only part of the story.

68%

Lower recorded storm fatality intensity, 1940-49 to 2010-19.*

438 / year

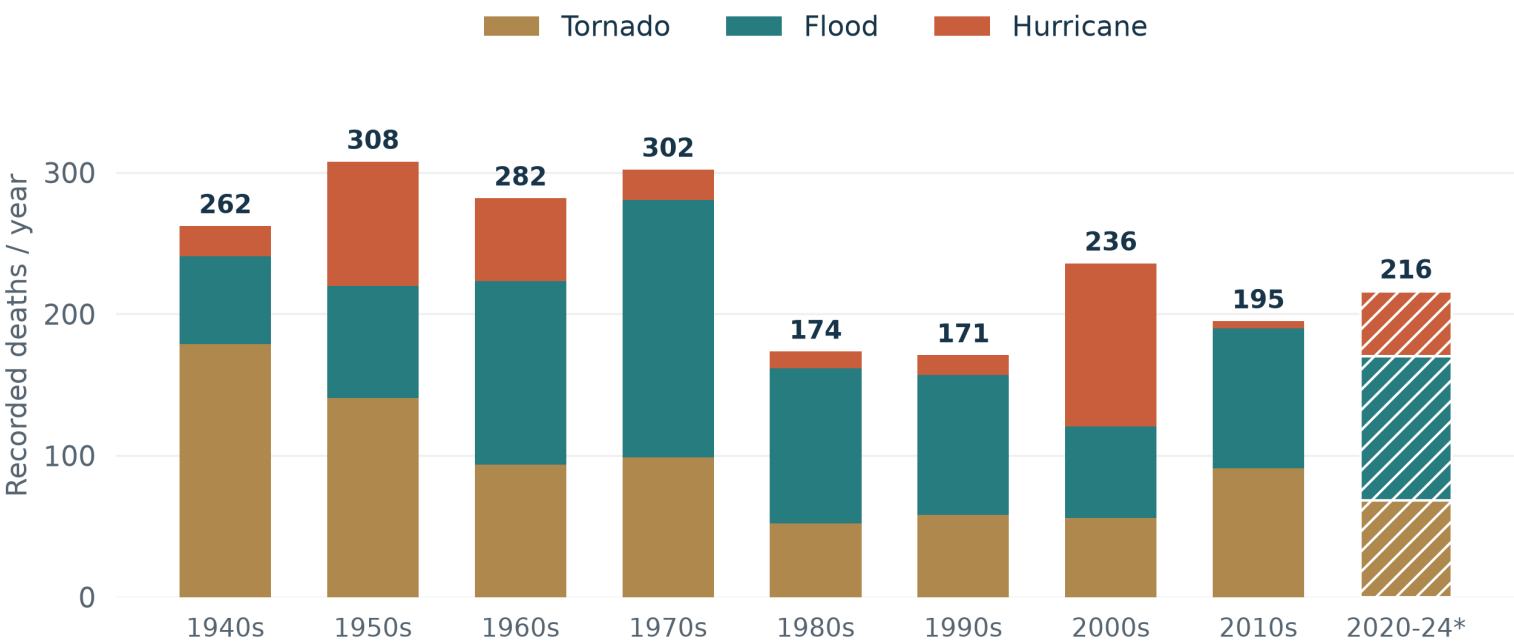
Recorded heat fatalities, 2020-24; 115 per year in 2010-19.

+21,267

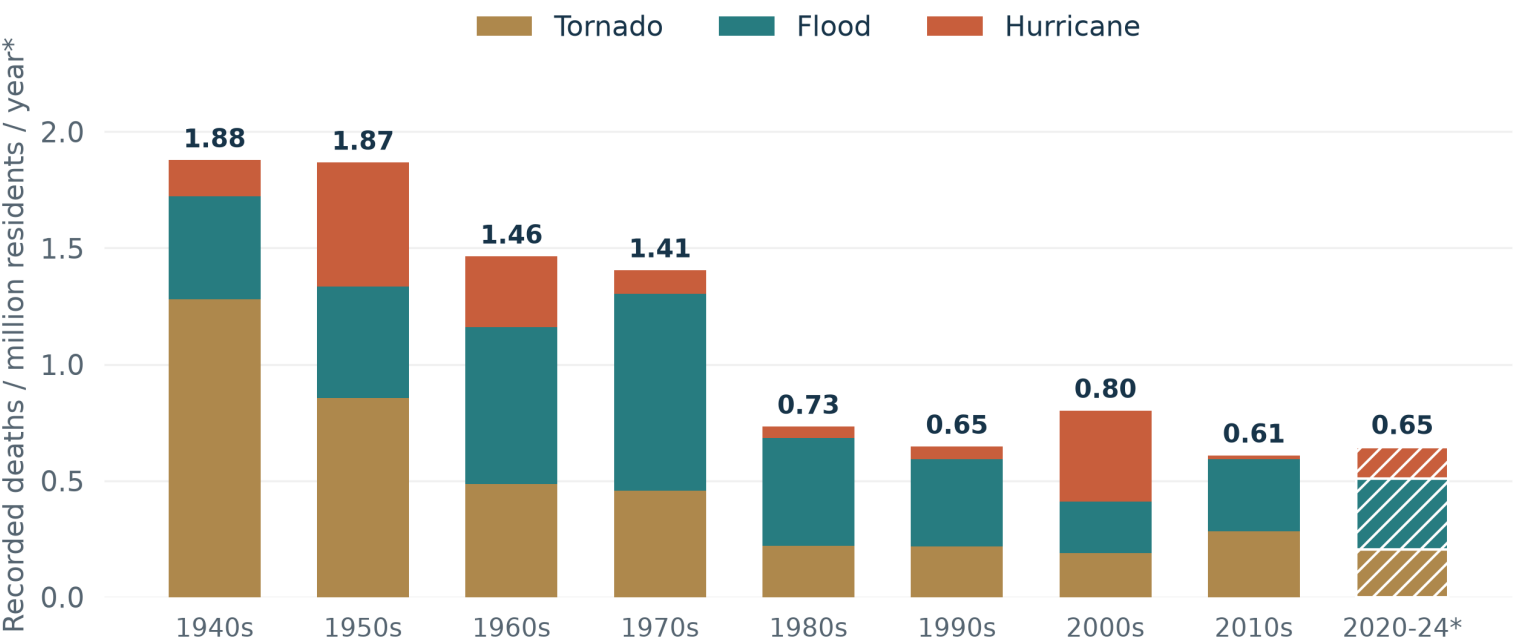
Net domestic migrants to Phoenix metro, July 2023-July 2024.

01 STORM DEATHS AND POPULATION

Annual average fatalities



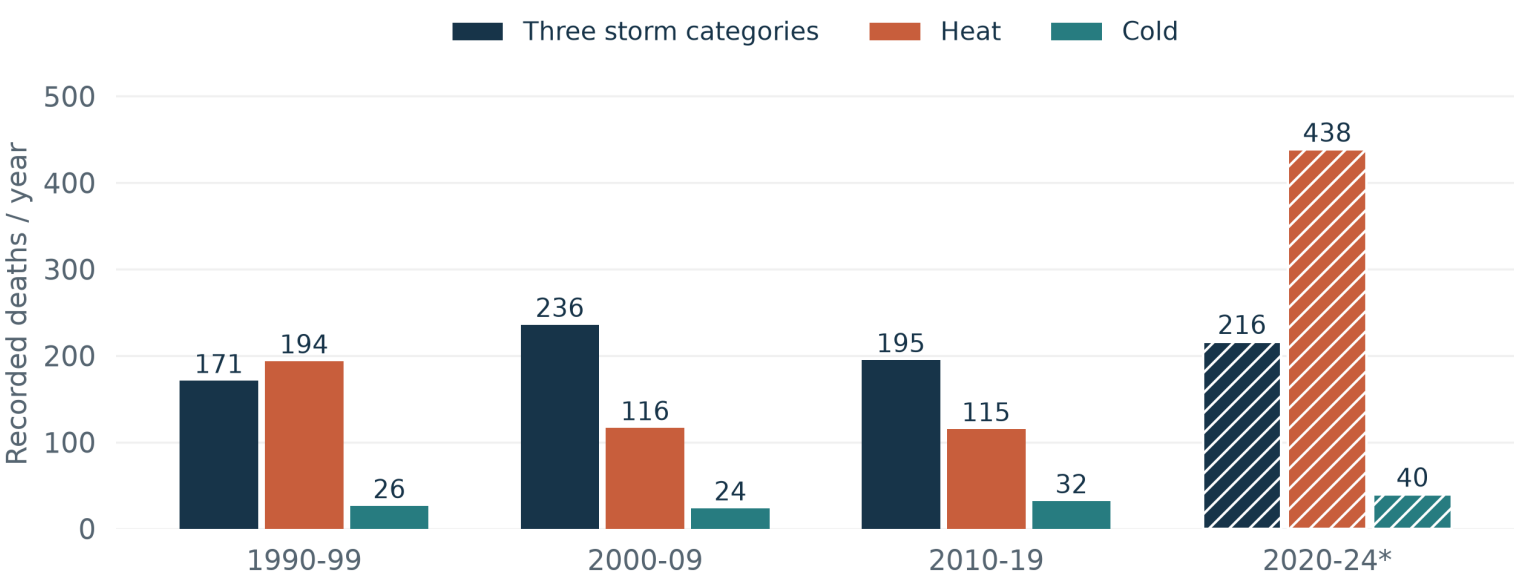
Population-adjusted intensity*



Tornado + flood + hurricane categories. Labels show totals. Full-period average (1940-2024): 239.8 deaths/year.

*NWS reporting includes territories; population covers 50 states + DC. This is an approximate indicator, not a precisely matched mortality rate. Hatched bars show five years (2020-24); other bars show ten. Reporting changes also matter.

02 RECORDED HEAT DEATHS HAVE RISEN



Heat and cold records begin in 1986 and 1988. Missing years are not zero. Cold excludes the separate winter-weather category.

WHY THE PAPER'S ESTIMATES ARE LARGER

NWS records deaths linked to weather events. Burke et al. ask how many deaths would be avoided if every day had the temperature with the lowest mortality. Their estimates also include effects on ordinary days and deaths recorded under other causes.

The two sources count different things. Neither directly measures deaths caused by recent climate change.

03 MOVING INTO THE HEAT

Net domestic migration, July 2023-July 2024

Maricopa County, AZ	-2,049
Pinal County, AZ	+23,316
Phoenix metro total	+21,267

Phoenix recorded 70 days at 110°F or hotter in 2024. The metro still gained domestic movers, but these totals do not reveal where they came from or how their personal exposure changed.

04 WOULD MORE AC HELP?

My first guess was that Europe had too little AC. I would still consider helping households pay for AC and electricity, but would compare the cost and health benefits with shade or better housing.

There is a problem with jumping straight to a subsidy: AC users may differ from other households in ways that also affect health. We need evidence of what the proposed program would achieve.

Sources: NWS, long-run fatalities & definitions: [weather.gov/hazstat](https://www.weather.gov/hazstat). Census, annual population and Vintage 2025 county migration: [census.gov/popest](https://www.census.gov/popest). NWS, 2024 Phoenix climate review: [weather.gov/psr/yearinreview2024](https://www.weather.gov/psr/yearinreview2024). Burke et al. (2025), NBER Working Paper 34313: [nber.org/papers/w34313](https://www.nber.org/papers/w34313). All source links and calculations are on the webpage.